

Marginal and Conditional Distribution

It focuses on a single variable's probabilities by summing over others in a joint distribution, appearing in table margins. = distribution of one variable while ignoring the other.

Example:

here we only care about the height

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_Y(y_j) = \sum_i p_{XY}(x_i, y_j)$$

and here, we only care about the age

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10
		1/10	2/10	2/10	0	3/10	2/10

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

$$p_X(x_i) = \sum_j p_{XY}(x_i, y_j)$$

$$p_X(7) = \sum_j p_{XY}(7, y_j)$$

$$p_X(7) = \frac{3}{10}$$

Conditional distribution shows the probability of one variable given a specific value of another, restricting focus to a subgroup.

Example:

		Height (Y)					
Age (X)		45	46	47	48	49	50
	9	0	0	0	0	3/10	1/10

Normalize

Divide by row sum

9	0	0	0	0	3/4	1/4
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$$P(Y = 49 | X = 9) = \frac{3}{4}$$

	Height (Y)						Sum
	45	46	47	48	49	50	P(X = 9)
(X)	9	0	0	0	0	3/10	1/10

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

$$p_{Y|X=9}(y) = P(Y = y | X = 9)$$

$$P(A, B) = P(A) \cdot P(B | A)$$

$$P(X = 9, Y = 49) = P(X = 9) \cdot P(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{P(X = 9, Y = 49)}{P(X = 9)}$$

Row sum

$$P(Y = 49 | X = 9) = \frac{3/10}{4/10}$$

$$p_{Y|X=9}(y) = P(Y = y | X = 9)$$

$$P(A, B) = P(A) \cdot P(B | A)$$

$$P(X = 9, Y = 49) = P(X = 9) \cdot P(Y = 49 | X = 9)$$

$$P(Y = 49 | X = 9) = \frac{P(X = 9, Y = 49)}{P(X = 9)}$$

Row sum

$$P(Y = 49 | X = 9) = \frac{3/10}{4/10} = \frac{3}{4}$$

Discrete Conditional Distribution

Formula

$$p_{Y|X=x}(y) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Conditional PDF of Y

Joint PDF of X and Y

Marginal distribution of X

Example:



Dice 1: 1/6 1/6 1/6 1/6 1/6 1/6

Dice 2: 1/6 1/6 1/6 1/6 1/6 1/6

$$\begin{aligned} p_{Y|X=4}(y = 1) &= \frac{p_{XY}(x = 4, y = 1)}{p_X(x = 4)} \\ &= \frac{1/36}{1/6} \\ &= \frac{1}{6} \end{aligned}$$

X

		Y					
	1	2	3	4	5	6	Sum
1	1/36	1/36	1/36	1/36	1/36	1/36	1/6
2	1/36	1/36	1/36	1/36	1/36	1/36	1/6
3	1/36	1/36	1/36	1/36	1/36	1/36	1/6
4	1/36	1/36	1/36	1/36	1/36	1/36	1/6
5	1/36	1/36	1/36	1/36	1/36	1/36	1/6
6	1/36	1/36	1/36	1/36	1/36	1/36	1/6